

2MVAa:Multivariable and Vector Analysis 1 Solutions to Exercise Sheet 2

Qu. 1. a) By considering the Taylor expansion for $1/(1-w)$ (about 0), where $|w| < 1$, show that

$$1/(1+t^2) = (1 - t^2 + t^4 - \dots).$$

Under what conditions is this expansion valid?

- b) By integrating this series term by term, find an expansion for $\arctan(u)$
 c) Find the Taylor expansion (up to and including cubic terms) of the function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ about the point $(1, 1, -1)$ where $f(x, y, z) = \arctan(x + y + 2z)$.

Solution. (a) Since $1/(1-w) = 1 + w + w^2 + w^3 + \dots$, putting $w = -t^2$ we obtain $1/(1+t^2) = 1 - t^2 + t^4 - t^6 + \dots$. This expansion is valid for $|t^2| < 1$, i.e. for $|t| < 1$.

(b) For $|u| < 1$,

$$\begin{aligned} u - \frac{u^3}{3} + \frac{u^5}{5} - \dots &= \int_0^u (1 - t^2 + t^4 - \dots) dt = \int_0^u \frac{dt}{1+t^2} dt \\ &= \tan^{-1} u - \tan^{-1} 0 = \tan^{-1} u = \arctan(u). \end{aligned}$$

(c) $f(x, y, z) = \tan^{-1}(x + y + 2z)$. For a Taylor expansion about the point $(1, 1, -1)$ we want an expansion in terms of $(x-1)$, $(y-1)$ and $[z - (-1)] = (z+1)$. Note that $x + y + 2z = (x-1) + (y-1) + 2(z+1)$, so, putting $u = [(x-1) + (y-1) + 2(z+1)]$, we have

$$\begin{aligned} f(x, y, z) &= \tan^{-1}(x + y + 2z) = \tan^{-1}[(x-1) + (y-1) + 2(z+1)] \\ &= [(x-1) + (y-1) + 2(z+1)] - \frac{[(x-1) + (y-1) + 2(z+1)]^3}{3} \\ &\quad + \frac{[(x-1) + (y-1) + 2(z+1)]^5}{5} - \dots \end{aligned}$$

Since we are only interested in terms that are cubic power or less we have

$$f(x, y, z) = [(x-1) + (y-1) + 2(z+1)] - \frac{[(x-1) + (y-1) + 2(z+1)]^3}{3} + \dots$$

or expanding (if you wish)

$$\begin{aligned} &= [(x-1) + (y-1) + 2(z+1)] \\ &\quad - \frac{1}{3} \left[(x-1)^3 + (y-1)^3 + 8(z+1)^3 + 12(x-1)(y-1)(z+1) \right. \\ &\quad \quad + 3(x-1)^2(y-1) + 6(x-1)^2(z+1) + 6(y-1)^2(z+1) \\ &\quad \quad \left. + 12(x-1)(z+1)^2 + 12(y-1)(z+1)^2 + 3(x-1)(y-1)^2 \right] + \dots \end{aligned}$$

Qu. 2. a) Given that the Taylor series for $\sin x$ about 0 is

$$\sum_{n \geq 0} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

find the Taylor series for $\sin(x+y)$ about $(0,0)$.

b) By using the fact that, for $|w| < 1$,

$$\frac{1}{1-w} = \sum_{n \geq 0} w^n,$$

find the Taylor series of $\frac{1}{1+\sin(x+y)}$ up to and including cubic terms.

Solution. a) We want an expansion for $\sin(x+y)$ in terms of $(x-0) = x$ and $(y-0) = y$, but then

$$\sin(x+y) = \sum_{n \geq 0} \frac{(-1)^n (x+y)^{2n+1}}{(2n+1)!}.$$

b) Putting $w = \sin(x+y)$ into $1/(1-w) = \sum_{n \geq 0} w^n$, when $|\sin(x+y)| < 1$ we have

$$\begin{aligned} \frac{1}{1+\sin(x+y)} &= \frac{1}{1-(-\sin(x+y))} = \frac{1}{1-(-(x+y) - (x+y)^3/3! + \dots)} \\ &= 1 - \left[(x+y) - \frac{1}{6}(x+y)^3 + \dots \right] + \left[(x+y) - \frac{1}{6}(x+y)^3 + \dots \right]^2 \\ &\quad - \left[(x+y) - \frac{1}{6}(x+y)^3 + \dots \right]^3 + \dots \end{aligned}$$

note that since we only want terms up to cubic powers, we do not need to take the expansion any further: all other terms are of higher order

$$\begin{aligned} &= 1 - [(x+y) - \frac{1}{6}(x+y)^3 + \dots] + [(x+y)^2 + \dots] - [(x+y)^3 + \dots] + \dots \\ &= 1 - (x+y) + (x+y)^2 - \frac{5}{6}(x+y)^3 \dots \quad \text{when } |\sin(x+y)| < 1 \end{aligned}$$

Qu. 3. Let P be the tangent plane to the surface $xyz = a^3$ at the point (r, s, t) . Find the Cartesian equation of P .

Let T be the tetrahedron formed by the coordinate planes, $x = 0$, $y = 0$, $z = 0$, and the tangent plane P . Show that the volume of T is independent of the point (r, s, t) . Recall that the volume of a tetrahedron is $\frac{1}{3}(\text{area of base} \times \text{height})$.

Solution. Suppose that (r, s, t) lies on the surface $xyz = a^3$. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the function $f(x, y, z) = xyz$, then the given surface is the level curve $f(x, y, z) = a^3$. $\nabla f(r, s, t) = st\mathbf{i} + tr\mathbf{j} + rs\mathbf{k}$. Remember that the equation of a tangent plane to a surface at a point with the position vector $\mathbf{a}$ is $\mathbf{r} \cdot \mathbf{n} = \mathbf{a} \cdot \mathbf{n}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ is the position vector of a point in this plane and $\mathbf{n}$ is a normal (not necessarily unit) vector

to the surface at point $\mathbf{a}$. After remembering this fact and also that the gradient is normal to the level curve (which is, in our case, a surface), we have that the equation of the tangent plane at (r, s, t) (i.e. for $\mathbf{a} = r\mathbf{i} + s\mathbf{j} + t\mathbf{k}$) is $stx + try + rsz = 3rst = 3a^3$ since $rst = a^3$ as (r, s, t) lies on the curve.

This plane intersects the x , y and z axes at the points $(3r, 0, 0)$, $(0, 3s, 0)$ and $(0, 0, 3t)$ (again noting $rst = a^3$). The volume of a tetrahedron is $\frac{1}{3} \cdot \text{area of base} \times \text{height}$, so our tetrahedron has volume

$$\frac{1}{3} \left(\frac{1}{2} \cdot 3r \cdot 3s \right) \cdot 3t = \frac{9}{2} rst = \frac{9}{2} a^3$$

which is independent of (r, s, t) as required.

Qu. 4. Find critical points of the following functions:

- a) $z = x^3 + y^3 - 3xy$ (determine the type of critical points)
- b) $z = xy\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$, $a > 0$, $b > 0$.

Solution.

- a) Conditions for critical points $\partial z / \partial x = 0 = \partial z / \partial y$ yield

$$3x^2 - 2y = 0, \quad 3y^2 - 3x = 0,$$

i.e. $x = y = 0$ and $x = y = 1$.

The Hessian matrix has the form

$$H = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix},$$

so that for the first point ($x = y = 0$) we have $H_1 = 0$, $H_2 = -9$ and hence, by the Leading Minor Test, it is a **saddle**.

For the second ($x = y = 1$) we have $H_1 = 6 > 0$, $H_2 = 27 > 0$ so that it is a **minimum**. (At this point $z = -1$.)

- b) Again, the conditions for the critical point give

$$y\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} - \frac{x^2 y}{a^2 \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}},$$

$$x\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} - \frac{xy^2}{b^2 \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}},$$

which make us notice that the expression under the square root should not be equal to zero and can be written down as

$$y \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right) - \frac{x^2 y}{a^2} = 0,$$

$$x \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2} \right) - \frac{xy^2}{b^2} = 0.$$

(A more convenient way of doing the calculations is to change the variables to $\bar{x} = x/a$, $\bar{y} = y/b$. Then you do not need to carry the coefficients with you all the way.)

Having in mind that $1 - x^2/a^2 - y^2/b^2 \neq 0$, we have that the solutions to these equations are

$$(1) \quad x = y = 0,$$

$$(2) \quad \frac{x}{a} = \frac{y}{b} = \pm \frac{1}{\sqrt{3}},$$

$$(3) \quad \frac{x}{a} = -\frac{y}{b} = \pm \frac{1}{\sqrt{3}}.$$

This is not part of what was for assessment but it is good for practice: After calculating the derivatives (which is again more convenient to do in terms of $\bar{x}$ and $\bar{y}$) and applying the Leading Minor Test, we find that (1) is a **saddle** (actually it was actually clear from the start since, to leading order as $x, y \rightarrow 0$, $z \approx xy$, which is a saddle), the roots (2) correspond to the local **maximums** and (3) to the local **minimums**. (Make sure that you do the calculations until the answers are correct!)

Qu. 5. Find the critical points of the function $f(x, y) = \sin x \sin y \sin(x + y)$ in an open domain $D = \{x, y : 0 < x < \pi, 0 < y < \pi\}$ and determine their type.

Solution. The critical points are determined by the equation $\nabla f = 0$.

For the given function one has

$$\nabla f = [\cos x \sin y \sin(x+y) + \sin x \sin y \cos(x+y)]\mathbf{i} + [\sin x \cos y \sin(x+y) + \sin x \sin y \cos(x+y)]\mathbf{j} = 0,$$

that is

$$(4) \quad f_x = \sin y \sin(2x + y) = 0$$

$$(5) \quad f_y = \sin x \sin(x + 2y) = 0$$

or, given that $x, y \neq 0, \pi$,

$$(6) \quad \sin(2x + y) = 0$$

$$(7) \quad \sin(x + 2y) = 0$$

The solution of the first equation gives a family of straight lines $2x + y = \pi n$, $n = 0, \pm 1, \pm 2, \dots$ of which only the lines $2x + y = \pi$ and $2x + y = 2\pi$ intersect with D .

The solution of the second equation are given by $x + 2y = \pi n$, $n = 0, \pm 1, \pm 2, \dots$ and only $x + 2y = \pi$ and $x + 2y = 2\pi$ intersect with D .

Thus one has 4 systems of linear algebraic equations:

$$\begin{array}{cccc} 2x + y = \pi & 2x + y = 2\pi & 2x + y = \pi & 2x + y = \pi \\ x + 2y = \pi & x + 2y = \pi & x + 2y = 2\pi & x + 2y = 2\pi \end{array},$$

Only the first and the third have solutions inside D defining a pair of critical points $x_1 = y_1 = \frac{\pi}{3}$ and $x_2 = y_2 = \frac{2\pi}{3}$.

The second derivatives are given by:

$$f_{xx} = 2 \sin y \cos(2x + y), \quad f_{xy} = \sin(2x + 2y), \quad f_{yy} = 2 \sin x \cos(x + 2y).$$

The Hessians at (x_1, y_1) and (x_2, y_2) have the form:

$$H(x_1, y_1) = \begin{pmatrix} -\sqrt{3} & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -\sqrt{3} \end{pmatrix} \quad \text{and} \quad H(x_2, y_2) = \begin{pmatrix} \sqrt{3} & \sqrt{3}/2 \\ \sqrt{3}/2 & \sqrt{3} \end{pmatrix}.$$

By the Leading Minor Test we have $\det H(x_1, y_1) = \det H(x_2, y_2) = 2\frac{1}{4} > 0$ so that neither point is a saddle. $f_{xx}(x_1, y_1) = -\sqrt{3} < 0$ so that (x_1, y_1) is a maximum and $f_{xx}(x_2, y_2) = \sqrt{3} > 0$ so that (x_2, y_2) is a minimum.
